

Supplementary Material for GRES: *Correlation-Aware One-Shot Pruning of LLMs across Granularities*

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This document accompanies the main paper, with material ordered as in the main text: Appendix A gives the method details, Appendix B the complete proofs, and Appendix C the additional experimental results and reproducibility details. Section, equation, and citation numbers are local to this document; cross-references to the main paper resolve automatically.

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A Method Details

This appendix expands the main text in order: the conceptual design-space figures and the complete method comparison, the standing assumptions and Gram shrinkage (main-text Section III-A), the detailed derivations

Table S1: **Full design space of one-shot LLM pruning**, with explicit pruning scores. Notation: θ_{ij} is a weight, $\mathbf{X}_j$ the j -th input channel, $\mathbf{G} = \mathbf{X}\mathbf{X}^\top$ the input Gram (layer Hessian), $\mathbf{H} = \mathbf{G} + \lambda \mathbf{I}$ its dampened form, g_{ij} a gradient/sensitivity term; for structured units $\mathbf{V}$ is the output-side projection ($\mathbf{W}_\downarrow$ for neurons, $\mathbf{W}_O$ for heads) and z_u the unit’s activation. RIA’s relative importance is $\text{RI}(\theta_{ij}) = \frac{|\theta_{ij}|}{\sum_k |\theta_{ik}|} + \frac{|\theta_{ij}|}{\sum_k |\theta_{kj}|}$. *Gran.*: U=unstructured, $N:M$ =semi-structured, S=structured. *Corr.*: correlation order across co-pruned units (none/diagonal/block/group/global). *Fwd*: criterion needs no back-propagation. *NoUpd*: retained weights unchanged. *1-shot*: no recovery fine-tuning. † GRES’s optional tier performs a block-local update.

Method	Pruning score σ_{ij} (unit score σ_u)	Gran.			Corr.	Fwd	NoUpd	1-shot
		U	$N:M$	S				
Magnitude [1]	$ \theta_{ij} $	✓	✓	×	none	✓	✓	✓
Wanda [2]	$ \theta_{ij} \ \mathbf{X}_j\ _2$	✓	✓	×	diag.	✓	✓	✓
RIA [3]	$\text{RI}(\theta_{ij}) \cdot \ \mathbf{X}_j\ _2^\alpha$	✓	✓	×	diag.	✓	✓	✓
GBLM-Pruner [4]	$(\theta_{ij} + \beta g_{ij}) \ \mathbf{X}_j\ _2$	✓	×	×	diag.	×	✓	✓
Pruner-Zero [5]	evolved $f(\theta_{ij} , \ \mathbf{X}_j\ _2, g_{ij})$	✓	×	×	diag.	×	✓	✓
RIA-permute [3]	RIA + input-channel permutation	×	✓	×	diag.	✓	✓	✓
SparseGPT [6]	$\theta_{ij}^2 / [\mathbf{H}^{-1}]_{jj}$	✓	✓	×	global	✓	×	✓
OBC [7]	$\theta_{ij}^2 / [\mathbf{H}^{-1}]_{jj}$ (exact OBS)	✓	×	✓	global	✓	×	✓
OBS [8]	$\theta_{ij}^2 / [\mathbf{H}^{-1}]_{jj}$ (full Hessian)	✓	×	×	global	×	×	✓
FLAP [9]	$\ \mathbf{V}[:, u]\ _2 \cdot \text{std}(z_u)$	×	×	✓	diag.	✓	×	✓
SliceGPT [10]	delete smallest PCA dims of $\mathbf{Z}\mathbf{Z}^\top$ (per-block)	×	×	✓	global	✓	×	✓
ZipLM [11]	$\theta_u^\top ([\mathbf{H}^{-1}]_{uu})^{-1} \theta_u$	×	×	✓	global	✓	×	×
LLM-Pruner [12]	$ \partial \mathcal{L} / \partial z_u \cdot z_u $ (grouped)	×	×	✓	group	×	×	×
GRES (ours)	$\mathbf{1}_P^\top \mathbf{\Gamma}_{PP} \mathbf{1}_P$, $\mathbf{\Gamma} = \mathbf{V}^\top \mathbf{V} \odot \mathbf{Z}\mathbf{Z}^\top$	✓	✓	✓	local	✓	✓ †	✓

of the saliency objectives, the efficient $N:M$ selection, and the fully detailed algorithm with its complexity analysis.

A.1 Full Design Space

This appendix gives the complete design-space comparison summarized in the main text, with the explicit pruning score of every method.

Table S2: Structured granularities under the unified Gram saliency (main-text Eq. (10)). Each unit removes the columns of an output map $\mathbf{V}$ scaled by a measured activation z_u ; the per-unit (diagonal) score is $\|\mathbf{V}[:, u]\|_2 \|z_u\|_2$, and the off-diagonal $\mathbf{V}^\top \mathbf{V} \odot \mathbf{Z}\mathbf{Z}^\top$ captures redundancy between co-pruned units. “Few” units (heads, GQA groups) admit *exact* correlation-aware selection; “many” units (neurons, value channels) use the diagonal rule or the greedy update (main-text Eq. (12)).

Structured unit	Output map $\mathbf{V}$	Activation z_u	Per-unit (diagonal) score	Units / layer $\rightarrow$ selection
MLP neuron	$\mathbf{W}_\downarrow$ (down-projection)	a_n (post-activation)	$\ \mathbf{W}_\downarrow[:, n]\ _2 \ a_n\ _2$	d_{ff} (many $\rightarrow$ diagonal / greedy)
Attention head	$\mathbf{W}_O[:, \mathcal{S}_h]$	$\mathbf{o}_h$ (head output)	$(\sum_t \ \mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)\ _2^2)^{1/2}$	tens (few $\rightarrow$ exact)
Attention value channel	$\mathbf{W}_O[:, (h, c)]$	$o_{h,c}$	$\ \mathbf{W}_O[:, (h, c)]\ _2 \ o_{h,c}\ _2$	d_{model} (many $\rightarrow$ diagonal / greedy)
Key-value head (GQA group)	$\{\mathbf{W}_O[:, \mathcal{S}_h]\}_{h \in g}$	$\{\mathbf{o}_h\}_{h \in g}$	group output energy	few groups (exact; shrinks KV cache)

Remark (layer Hessian vs. task Fisher). The quantity $\|\mathbf{X}_j\|_2^2 = \mathbf{G}_{jj}$ is the diagonal of the layer-reconstruction Hessian $\mathbf{G}$, not of the task-loss Fisher. We deliberately build the criterion on the reconstruction objective, for which $\mathbf{G}$ is the *exact* second-order object and no scaling exponent is needed. Reweighting $\|\mathbf{X}_j\|_2$ by a learned power (as in activation-reweighted variants) is recovered as a diagonal special case but is orthogonal to, and dominated by, the correlation correction GRES introduces.

A.2 Standing Assumptions and Gram Shrinkage

This appendix expands the two preliminaries summarized in the main text (Section III-A): the standing assumptions used throughout, and the properties of the Gram shrinkage (3).

Standing assumptions. We make three assumptions, used throughout and revisited where they bind. **(A1)** The calibration activations are fixed: when scoring a layer, the inputs $\mathbf{X}$ (and hence $\mathbf{G}$) are those induced by the already-pruned preceding layers, and removing units *within* the current layer does not alter $\mathbf{X}$. **(A2)** $\mathbf{G} = \mathbf{X}\mathbf{X}^\top \succeq 0$ is positive semidefinite; for the optional compensation of Section III-C we additionally assume the relevant block $\mathbf{G}_{BB} \succ 0$, which is ensured in practice by the standard Hessian dampening $\mathbf{G} \leftarrow \mathbf{G} + \lambda \mathbf{I}$, $\lambda > 0$ [6]. **(A3)** Within a layer, the removal directions of distinct units are *additive and mutually non-interacting*: removing a set P perturbs the output by exactly $\sum_{u \in P} \delta_u$. (A3) holds for the units we prune—individual weights, MLP neurons, and attention heads—because each neuron’s pre-output activation and each head’s output are computed independently of the others from the shared layer input; it is what makes the saliency *exact* rather than a first-order surrogate (see Appendix A.3.1).

Gram shrinkage. The criterion uses the off-diagonal entries of $\mathbf{G}$, which are precisely the entries most corrupted by the finite calibration set: with T tokens ($\sim 10^4$ – 10^5) estimating an $i \times i$ second-moment matrix ($i \sim 10^4$), the empirical $\mathbf{G} = \mathbf{X}\mathbf{X}^\top$ is a noisy estimate of the population Gram, and the noise concentrates off the diagonal. Since GRES’s departure from Wanda lives entirely in those off-diagonals, we regularize them with the Ledoit–Wolf-style shrinkage toward the (well-estimated) diagonal of Eq. (3), $\hat{\mathbf{G}} = (1 - \rho) \mathbf{G} + \rho \text{diag}(\mathbf{G})$ with $\rho \in [0, 1]$, and use $\hat{\mathbf{G}}$ wherever $\mathbf{G}$ appears in selection and compensation. The shrinkage intensity ρ admits the closed-form Ledoit–Wolf estimator (the ratio of the off-diagonal sampling variance to the off-diagonal energy), so it is computed from the same calibration statistics with no tuning. Three properties make this a strict, low-cost improvement rather than a new mechanism. (i) It preserves the diagonal exactly, so the Wanda corner—and Proposition 1—is untouched: at $\rho=1$, GRES reduces to Wanda. (ii) It interpolates monotonically between correlation-aware ($\rho=0$) and diagonal ($\rho=1$) selection, so it can only *denoise*, never invert, the contribution of the off-diagonal block; the suboptimality bound of Theorem 1 applies to $\hat{\mathbf{G}}$ verbatim with $\text{Off}(P)$ scaled by $(1 - \rho)$. (iii) Because $\text{diag}(\mathbf{G}) \succeq 0$ and $\mathbf{G} \succeq 0$, the convex combination keeps $\hat{\mathbf{G}} \succeq 0$, and for $\rho > 0$ it is strictly diagonally loaded, so $\hat{\mathbf{G}}_{BB} \succ 0$ automatically—absorbing the dampening of (A2) without a separately tuned λ .

A.3 Detailed Derivations

Throughout, $\mathbf{X} \in \mathbb{R}^{i \times T}$ are the (fixed, by (A1)) calibration activations of the layer, $\mathbf{G} = \mathbf{X}\mathbf{X}^\top$, and for a block/index set B we write $\mathbf{X}_B$ for the corresponding rows of $\mathbf{X}$ and $\mathbf{G}_{BB} = \mathbf{X}_B \mathbf{X}_B^\top$.

A.3.1 The Gram saliency (6)

Let P be a set of units with removal directions $\delta_u(t)$. By assumption (A3) the units’ contributions are additive and do not interact, so deleting all of P changes the layer output at token t by exactly $\Delta(t) = -\sum_{u \in P} \delta_u(t)$. The reconstruction error is the total squared deviation,

$$\begin{aligned} \text{Sal}(P) &= \sum_{t=1}^T \|\Delta(t)\|_2^2 = \sum_{t=1}^T \left\langle \sum_{u \in P} \delta_u(t), \sum_{v \in P} \delta_v(t) \right\rangle = \sum_{t=1}^T \sum_{u \in P} \sum_{v \in P} \langle \delta_u(t), \delta_v(t) \rangle \\ &= \sum_{u \in P} \sum_{v \in P} \underbrace{\sum_{t=1}^T \langle \delta_u(t), \delta_v(t) \rangle}_{=\Gamma_{uv}} = \sum_{u,v \in P} \Gamma_{uv} = \mathbf{1}_P^\top \mathbf{\Gamma}_{PP} \mathbf{1}_P, \end{aligned}$$

where the interchange of finite sums is justified termwise and $\mathbf{1}_P \in \mathbb{R}^{|P|}$ is all-ones. The diagonal part is $\sum_{u \in P} \Gamma_{uu} = \sum_{u \in P} \sum_t \|\delta_u(t)\|_2^2$, i.e. the sum of per-unit removal energies; in every instance below this energy factors into a weight-side norm times an activation-side norm, which is the source of the Wanda-style diagonal score.

The only place an assumption enters is the first equality $\Delta(t) = -\sum_{u \in P} \delta_u(t)$, i.e. that the joint perturbation is the sum of the individual ones. This is (A3), and it holds *exactly*—not merely to first order—for the units we prune, because each is a distinct, linearly contributing term of the layer output: for weights of a row, the output $\sum_j \theta_{rj} \mathbf{X}_{jt}$ is linear in the weights, so removing several entries subtracts their contributions additively; for MLP neurons and attention heads/value channels, the output $\sum_n \mathbf{W}_\downarrow[:, n] a_n(t)$ (resp. $\sum_h \mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)$) is a sum over units whose individual terms are computed independently of one another from the shared layer input, so zeroing a subset removes exactly those terms and leaves the rest unchanged. Consequently $\text{Sal}(P)$ is the *exact* layer reconstruction error of removing P within the layer, with no Taylor remainder; the only modeling assumption is (A1) (the input $\mathbf{X}$ is held at its already-pruned value), which is the standard one-shot layer-wise convention shared by Wanda and SparseGPT [2, 6].

A.3.2 Unstructured exact error (5)

A single weight θ_{rj} (row r , input j) contributes the scalar $\theta_{rj} \mathbf{X}_{jt}$ to the scalar output $y_r(t) = \sum_j \theta_{rj} \mathbf{X}_{jt}$, so its removal direction is $\delta_j(t) = \theta_{rj} \mathbf{X}_{jt}$. Substituting into the definition of Γ in §A.3.1,

$$\Gamma_{jk} = \sum_{t=1}^T \delta_j(t) \delta_k(t) = \theta_{rj} \theta_{rk} \sum_{t=1}^T \mathbf{X}_{jt} \mathbf{X}_{kt} = \theta_{rj} \theta_{rk} \mathbf{G}_{jk}, \quad \text{i.e.} \quad \Gamma = \boldsymbol{\theta}_r \boldsymbol{\theta}_r^\top \odot \mathbf{G}.$$

Hence $\text{Sal}(P) = \mathbf{1}_P^\top \Gamma_{PP} \mathbf{1}_P = \sum_{j,k \in P} \theta_{rj} \theta_{rk} \mathbf{G}_{jk} = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P}$, which is (5); this agrees with the residual-norm computation in the proof of Proposition 1, providing a consistency check. The diagonal term is $\Gamma_{jj} = \theta_{rj}^2 \mathbf{G}_{jj} = (|\theta_{rj}| \|\mathbf{X}_j\|_2)^2 = \sigma_{rj}^2$.

A.3.3 Block-local Optimal-Brain-Surgeon update (9)

Fix row r and block B with $\mathbf{G}_{BB} \succ 0$ (A2). Prune $P \subseteq B$ and allow the retained weights $K = B \setminus P$ to change; write the change to the block weights as $\Delta \in \mathbb{R}^{|B|}$. Since only block weights change, the deviation of the row output from its dense value is $\Delta^\top \mathbf{X}_B$, so the block reconstruction error is

$$f(\Delta) = \|\Delta^\top \mathbf{X}_B\|_2^2 = \Delta^\top \mathbf{G}_{BB} \Delta.$$

Pruning forces the pruned coordinates to zero, i.e. $\Delta_P = -\boldsymbol{\theta}_{r,P}$, while Δ_K is free. Partitioning $\mathbf{G}_{BB} = \begin{pmatrix} \mathbf{G}_{PP} & \mathbf{G}_{PK} \\ \mathbf{G}_{KP} & \mathbf{G}_{KK} \end{pmatrix}$,

$$f(\Delta) = \Delta_P^\top \mathbf{G}_{PP} \Delta_P + 2\Delta_P^\top \mathbf{G}_{PK} \Delta_K + \Delta_K^\top \mathbf{G}_{KK} \Delta_K.$$

This is strictly convex in Δ_K (as $\mathbf{G}_{KK} \succ 0$); setting $\nabla_{\Delta_K} f = 2\mathbf{G}_{KP} \Delta_P + 2\mathbf{G}_{KK} \Delta_K = 0$ gives the unique minimizer

$$\Delta_K^* = -\mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \Delta_P = \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P}.$$

Thus the retained weights are updated by $\boldsymbol{\theta}_{r,K} \leftarrow \boldsymbol{\theta}_{r,K} + \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P}$. To match the form in (9), use the block-inverse identities of Lemma 1(i)–(ii) (with Schur complement $\mathbf{S}_P = \mathbf{G}_{PP} - \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP}$),

$$[\mathbf{G}_{BB}^{-1}]_{PP} = \mathbf{S}_P^{-1}, \quad [\mathbf{G}_{BB}^{-1}]_{KP} = -\mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \mathbf{S}_P^{-1},$$

so that $[\mathbf{G}_{BB}^{-1}]_{KP} ([\mathbf{G}_{BB}^{-1}]_{PP})^{-1} = -\mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \mathbf{S}_P^{-1} \mathbf{S}_P = -\mathbf{G}_{KK}^{-1} \mathbf{G}_{KP}$, and the update equals $\boldsymbol{\theta}_{r,K} \leftarrow -[\mathbf{G}_{BB}^{-1}]_{KP} ([\mathbf{G}_{BB}^{-1}]_{PP})^{-1} \boldsymbol{\theta}_{r,P}$, as claimed. Substituting Δ_K^* back, the minimal error collapses to the Schur-complement form,

$$\begin{aligned} f(\Delta^*) &= \Delta_P^\top \mathbf{G}_{PP} \Delta_P - 2\Delta_P^\top \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \Delta_P + \Delta_P^\top \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \Delta_P \\ &= \Delta_P^\top (\mathbf{G}_{PP} - \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP}) \Delta_P = \boldsymbol{\theta}_{r,P}^\top \mathbf{S}_P \boldsymbol{\theta}_{r,P} = \boldsymbol{\theta}_{r,P}^\top ([\mathbf{G}_{BB}^{-1}]_{PP})^{-1} \boldsymbol{\theta}_{r,P}, \end{aligned}$$

which is $\text{Sal}_{\text{comp}}(P)$ in (9). Setting $\Delta_K = 0$ (no compensation) gives instead $f = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P}$, the uncompensated cost (7); Proposition 2 compares the two.

A.3.4 Structured saliencies (11) and its variants

MLP neurons. Neuron n contributes $\mathbf{W}_\downarrow[:, n] a_n(t) \in \mathbb{R}^{d_{\text{model}}}$ to the block output $\mathbf{W}_\downarrow \mathbf{a}(t)$, and by (A3) the activations a_m , $m \neq n$, are unaffected by deleting neuron n ; hence $\delta_n(t) = \mathbf{W}_\downarrow[:, n] a_n(t)$. Then

$$\Gamma_{nm} = \sum_{t=1}^T \langle \mathbf{W}_\downarrow[:, n] a_n(t), \mathbf{W}_\downarrow[:, m] a_m(t) \rangle = \underbrace{(\mathbf{W}_\downarrow[:, n])^\top \mathbf{W}_\downarrow[:, m]}_{=\mathbf{M}_{nm}} \underbrace{\sum_{t=1}^T a_n(t) a_m(t)}_{=\mathbf{C}_{nm}} = (\mathbf{M} \odot \mathbf{C})_{nm},$$

giving $\text{Sal}(P) = \mathbf{1}_P^\top (\mathbf{M} \odot \mathbf{C})_{PP} \mathbf{1}_P$, which is (11). The diagonal is $\Gamma_{nn} = \|\mathbf{W}_\downarrow[:, n]\|_2^2 \|a_n\|_2^2$, whose square root is the per-neuron score $\|\mathbf{W}_\downarrow[:, n]\|_2 \cdot \|a_n\|_2$.

Attention value channels. A value channel (h, c) contributes the single column $\mathbf{W}_O[:, (h, c)] o_{h,c}(t)$ to the residual—an MLP neuron in attention space—so it is the neuron derivation with $\mathbf{W}_\downarrow \rightarrow \mathbf{W}_O$ and $\mathbf{a} \rightarrow \mathbf{o}$: the diagonal score is $\|\mathbf{W}_O[:, (h, c)]\|_2 \cdot \|o_{h,c}\|_2$, with the same $\mathbf{W}_O$ -gram $\odot$ activation-gram off-diagonal correction. The attention granularities thus nest (value channels $\subset$ heads) exactly as the weight ones do, with whole-head removal recovered by grouping a head’s d_h channels (below). We derive this granularity for completeness but do not evaluate it experimentally.

Attention heads. Head h contributes $\mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)$, so $\delta_h(t) = \mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)$ and

$$\Gamma_{hh'} = \sum_{t=1}^T \mathbf{o}_h(t)^\top \mathbf{W}_O[:, \mathcal{S}_h]^\top \mathbf{W}_O[:, \mathcal{S}_{h'}] \mathbf{o}_{h'}(t), \quad \Gamma_{hh} = \sum_{t=1}^T \|\mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)\|_2^2,$$

the head’s output energy. Grouping a head’s d_h value channels recovers this Γ_{hh} , confirming the channel $\subset$ head nesting.

Key–value (GQA) groups. A group g of query heads sharing one key–value head is removed together, so $\delta_g(t) = \sum_{h \in g} \mathbf{W}_O[:, \mathcal{S}_h] \mathbf{o}_h(t)$ and $\Gamma_{gg'} = \sum_t \langle \delta_g(t), \delta_{g'}(t) \rangle$; this is the head form with the comparison group widened from a head to a GQA group. Equivalently, the group saliency aggregates the head-level Gram of Eq. (10) over the query heads in the group,

$$\text{score}(g) = \sum_{h, h' \in g} \Gamma_{hh'},$$

and GRES-group selects KV groups by greedy marginal-cost removal on this group Gram (Eq. (12) with units = groups). Concretely, in Mistral-7B (32 query heads, 8 KV heads) each KV group holds 4 query heads, so removing one group deletes 4 query heads and 1 KV head at once and directly shrinks the KV cache. Because a removal step deletes a *whole* KV group, the granularity GRES-group can express is set by the number of KV groups (Table S3): with ≥ 4 groups a step is fine-grained enough for both ratios, whereas Qwen2.5-7B’s four groups make the 20% target round to zero groups (no pruning, marked “*”), and Qwen2.5-3B’s two groups leave no valid operation—at 20% the minimum removal (one group) is already 50%, and at 50% removing one of two groups collapses GQA from two groups to one, halving the attention capacity in a single destructive step (marked “N/A[†]”). GRES-group is therefore the recommended variant for GQA models with ≥ 8 KV groups (LLaMA-3.2-3B, Mistral-7B), while GRES-head (operating at the finer 16-head granularity) is preferred for Qwen2.5-3B.

A.4 Channel Permutation Procedures

Equation (8) of the main text defines the ideal permutation objective: the assignment of input channels to $N:M$ blocks that minimizes the total post-selection block error. Solving it exactly is a balanced graph-partitioning problem, so we replace it by a surrogate computed once per layer from the shrunk Gram $\hat{\mathbf{G}}$ and shared across all o output rows. We state both surrogates used in the paper; the cheap variant (v2) is the default throughout, and main-text Table III compares them.

Functional equivalence. A permutation of the input channels is undone by the same permutation applied upstream: with $\mathbf{P}_\pi$ the permutation matrix, $\Theta \mathbf{X} = (\Theta \mathbf{P}_\pi)(\mathbf{P}_\pi^\top \mathbf{X})$, so we permute the weight columns once,

Table S3: Why GRES-group’s applicability depends on the KV-group count: a removal step deletes one whole KV group.

Model	KV groups	20% $\rightarrow$ remove	50% $\rightarrow$ remove
LLaMA-2-7B/13B	32 / 40 (MHA)	fine (= head)	fine (= head)
Mistral-7B	8	~ 2 groups (25%)	4 groups (50%)
LLaMA-3.2-3B	8	~ 2 groups (25%)	4 groups (50%)
Qwen2.5-7B	4	0 groups (* no-prune)	2 groups (50%)
Qwen2.5-3B	2	impossible (min 50%)	too coarse (1 of 2)

select and prune in the permuted order, and either fold $\mathbf{P}_\pi^\top$ into the preceding projection or store π as an index vector applied to the layer input. The pruned layer computes the same function as the unpermuted layer carrying the same mask, the $N:M$ block structure is unchanged, and no additional runtime arithmetic is incurred.

Algorithm S1 Cheap permutation (v2) — the default

Require: shrunk Gram diagonal $\widehat{\mathbf{G}}_{jj}$; weights $\boldsymbol{\Theta} \in \mathbb{R}^{o \times i}$; block size M ; $b = i/M$ blocks

- 1: $s_j \leftarrow \sqrt{\widehat{\mathbf{G}}_{jj}} \|\boldsymbol{\Theta}_{:,j}\|_2$ for $j = 1, \dots, i$ $\triangleright$ row-aggregated channel importance
 - 2: $\sigma \leftarrow \text{argsort}(s)$ in decreasing order $\triangleright O(i \log i)$
 - 3: **for** $t = 0, \dots, i - 1$ **do**
 - 4: assign channel $\sigma[t]$ to block $t \bmod b$ $\triangleright$ round-robin deal
 - 5: **end for**
 - 6: $\pi \leftarrow$ concatenation of the b blocks in index order
 - 7: **return** π $\triangleright$ apply to weight columns; then select by (7) within each block
-

Algorithm S2 Spectral permutation — the alternative of Table III

Require: shrunk Gram $\widehat{\mathbf{G}}$; block size M ; $b = i/M$ blocks

- 1: $A_{jk} \leftarrow |\widehat{\mathbf{G}}_{jk}| / \sqrt{\widehat{\mathbf{G}}_{jj} \widehat{\mathbf{G}}_{kk}}$ $\triangleright$ normalized affinity
 - 2: embed the channels in the leading eigenvectors of the normalized affinity $\triangleright O(i^3)$ worst case
 - 3: partition the embedded channels into b groups of exactly M channels
 - 4: $\pi \leftarrow$ concatenation of the groups
 - 5: **return** π
-

What the two surrogates encode. They approximate (8) from opposite directions. The spectral surrogate groups channels that co-activate, so that each block carries as much within-block structure as possible for the off-diagonal criterion (7) to exploit. The cheap surrogate equalizes the importance mass across blocks, so that no single block is forced to discard several of its most important channels—the effect illustrated in main-text Figure 2. That figure depicts a single output row for readability; because one ordering must serve all o rows, Algorithm S1 ranks channels by the row-aggregated statistic $\sqrt{\widehat{\mathbf{G}}_{jj}} \|\boldsymbol{\Theta}_{:,j}\|_2$, whose per-row counterpart $\sqrt{\widehat{\mathbf{G}}_{jj}} |\theta_{rj}|$ is the bar height in the figure. Both are computed once per layer and reused for every output row; the cheap surrogate costs $O(i \log i)$ against the $O(i^3)$ worst case of the eigendecomposition, which is also why it scales to the 18,944-wide layers of Qwen2.5-7B. The identity ordering is feasible in (8), so the exact minimizer of that objective can only lower the attained block error; the surrogates approximate that minimizer at negligible cost, and their measured effect is reported in main-text Table III and Table S16.

A.5 Efficient $N:M$ Selection

The exact $N:M$ selection of Eq. (7) evaluates, per output row r and block B , the $\binom{M}{N}$ candidate quadratic forms $\theta_{r,P}^\top \hat{\mathbf{G}}_{BB,PP} \theta_{r,P}$ and takes the minimizer. Its arithmetic cost is modest— $O(o \binom{M}{N} (M-N)^2/M)$ per layer, with $\hat{\mathbf{G}}_{BB}$ shared across all o rows—but a naive loop over rows, blocks and patterns is dominated by interpreter and kernel-launch overhead. Three implementation techniques remove it. Batched enumeration (1) and Wanda-bound block skipping (2) are *exact*: they never alter a selected mask. Mixed-precision block statistics (3) alters a mask only when a runner-up margin falls below the bf16 rounding error—exactly the near-ties that (2) already diverts to the full fp32 search. Throughout, i is the input width, o the number of rows, M the block size, and $b = i/M$ the blocks per row.

(1) Batched enumeration as dense tensor contractions. The set of admissible keep/prune patterns is *identical* for every block of every row: there are exactly $\binom{M}{N}$ of them, fixed by (N, M) . We precompute the prune-supports as a constant index tensor and replace the per-block loop by a single batched contraction over the whole layer. Reshape the row weights into blocks $\Theta \in \mathbb{R}^{o \times b \times M}$ and gather the per-block grams into $\hat{\mathbf{G}}_{\text{blk}} \in \mathbb{R}^{b \times M \times M}$ (one $M \times M$ matrix per block, formed once and shared across rows). For each of the $K = \binom{M}{N}$ prune-patterns p with support $P_p \subseteq \{1, \dots, M\}$, $|P_p| = M - N$, the cost of pruning P_p in every (row, block) simultaneously is the tensor

$$\mathbf{C}_p[r, c] = \sum_{j, k \in P_p} \Theta[r, c, j] \hat{\mathbf{G}}_{\text{blk}}[c, j, k] \Theta[r, c, k] = (\Theta_{P_p} \cdot (\hat{\mathbf{G}}_{\text{blk}} \Theta_{P_p})) [r, c], \quad (1)$$

a single batched matrix–vector product followed by a contraction, evaluated for all ob blocks at once; stacking $\{\mathbf{C}_p\}_{p=1}^K$ and taking $\arg \min_p$ along the pattern axis yields every block’s optimal mask in K dense kernels per layer (e.g. $K = 6$ for 2:4, $K = 70$ for 4:8). The arithmetic is unchanged from Eq. (7); the gain is the elimination of $O(ob)$ host-side iterations in favor of $O(K)$ GPU kernels, which on modern accelerators reduces wall-clock by orders of magnitude and is what makes the method competitive with the fused baselines.

(2) Wanda-bound block skipping. Most blocks are decided identically by the diagonal (Wanda) rule and by the exact search, so the exact search is wasted on them. Theorem 1 makes this checkable *a priori*: for a block B , write $\text{Diag}(P) = \sum_{j \in P} \sigma_{rj}^2$ and $\text{Off}(B) = \sum_{j \neq k \in B} |\theta_{rj} \theta_{rk}| |\hat{\mathbf{G}}_{jk}|$ (an upper bound on the off-diagonal contribution of *any* pattern in the block). Let $P_W = \arg \min_{|P|=M-N} \text{Diag}(P)$ be Wanda’s pick and P_2 the runner-up by Diag . By the sandwich $\text{Diag}(P) - \text{Off}(B) \leq \text{Sal}(P) \leq \text{Diag}(P) + \text{Off}(B)$ of Theorem 1, if

$$\text{Diag}(P_2) - \text{Diag}(P_W) > 2 \text{Off}(B), \quad (2)$$

then $\text{Sal}(P_W) < \text{Sal}(P)$ for every other pattern P , so P_W is *provably* the exact minimizer and the block needs no search. Computing Diag and $\text{Off}(B)$ is $O(M^2)$ from the already-formed $\hat{\mathbf{G}}_{BB}$ and is itself batched across the layer; blocks passing the test (2) take the Wanda mask directly, and only the remaining fraction α (those with large relative off-diagonal mass, i.e. large ρ_B) enter the batched enumeration of technique (1). Because the bound is sufficient, this is *exact*: it never changes a selected mask, it only avoids redundant work. Empirically α is well below 1—consistent with Proposition 3, where the correlation correction is governed by ρ_B and is negligible when ρ_B is small—so the average per-block cost approaches Wanda’s on the easy majority of blocks and incurs the full search only where it can matter. With shrinkage (3) the $\text{Off}(B)$ term scales by $(1 - \rho)$, so stronger shrinkage makes more blocks skippable, smoothly trading a (provably bounded) amount of search for speed.

(3) Mixed-precision block statistics. The block grams $\hat{\mathbf{G}}_{BB}$ enter selection only through a ranking—the $\arg \min$ over $\binom{M}{N}$ candidate costs—which is invariant to small relative perturbations of the costs. We therefore store and contract the block grams in bf16, reserving fp32 only for the optional compensation tier (9), whose Cholesky/inverse *is* precision-sensitive. This halves the memory traffic of the dominant operation (1) (small batched matmuls are bandwidth-bound, so the wall-clock saving is close to the bandwidth saving) and leaves the selected masks unchanged whenever the runner-up margin exceeds the bf16 rounding error—the same margin that technique (2) already computes, so the two compose: blocks with a margin too small for bf16

Algorithm S3 GRES: one-shot, layer-wise pruning (full listing of the sweep described at the end of main-text Section III-E)

Require: calibration set $\mathcal{D}$; granularity $g \in \{\text{unstruct}, N:M, \text{struct}\}$; target sparsity; compensation flag c ; permutation flag p

- 1: **for** each layer ℓ (loaded to GPU in order) **do**
- 2: forward $\mathcal{D}$ through ℓ and accumulate statistics:
- 3: $g = \text{unstruct}$: $\mathbf{G}_{jj} = \|\mathbf{X}_j\|_2^2$ $\triangleright$ Wanda diagonal
- 4: $g = N:M$: block grams $\{\mathbf{G}_{BB}\}$ $\triangleright O(iM)$, shared over rows
- 5: $g = \text{struct}$: neuron energies $\|a_n\|_2$ and/or head energies; weight grams $\mathbf{M}$
- 6: shrink: $\hat{\mathbf{G}} \leftarrow (1 - \rho)\mathbf{G} + \rho \text{diag}(\mathbf{G})$, ρ by Ledoit–Wolf $\triangleright$ denoise off-diagonals (3)
- 7: **if** p and $g = N:M$ **then** build π from $\hat{\mathbf{G}}$ by Algorithm S1 (default) or S2 $\triangleright$ once per layer, shared over rows
- 8: **end if**
- 9: **for** each comparison group (output row / block / unit set) **do**
- 10: select pruned subset by minimizing $\text{Sal}(\cdot)$ using $\hat{\mathbf{G}}$:
- 11: unstruct: threshold smallest $|\theta_{rj}| \|\mathbf{X}_j\|_2$
- 12: $N:M$: $P_r^*(B) = \arg \min_{|P|=M-N} \boldsymbol{\theta}_{r,P}^\top \hat{\mathbf{G}}_{BB,PP} \boldsymbol{\theta}_{r,P}$ $\triangleright$ enumerate $\binom{M}{M-N}$
- 13: struct: rank units by Sal , remove smallest to target ratio
- 14: **end for**
- 15: **if** c **then** apply block-local compensation (9)
- 16: **end if**
- 17: write back mask (and updates; undo π upstream), forward pruned activations to $\ell+1$, free ℓ
- 18: **end for**

to resolve are exactly the near-ties that technique (2) would route to the (fp32) full search anyway. We use bf16 rather than fp16 because shrinkage (3) multiplies the off-diagonals by $(1 - \rho)$, and bf16’s wider exponent preserves their dynamic range.

Combined cost. Writing c_W for Wanda’s per-layer cost (one fused magnitude $\times$ norm pass and a per-block top- N), the combined selection cost is approximately

$$c_W + \underbrace{O(oiM)}_{\text{Diag, Off test, batched, bf16}} + \alpha \cdot \underbrace{O(oi \binom{M}{N} (M-N)^2 / M)}_{\text{full search on hard blocks only}},$$

with $\alpha \ll 1$ in practice and every term a batched GPU kernel. This keeps the no-compensation variant within a small constant of Wanda and well below SparseGPT, which pays an $O(d_{\text{hidden}}^3)$ inverse and a weight-update pass that none of the above incurs. The compensation tier, when enabled, adds the $M \times M$ block inverses of Eq. (9) in fp32, still independent of d_{hidden} and shared across rows.

A.6 Full Pruning Algorithm and Complexity Analysis

Complexity analysis. Let a layer have o output rows, i input channels, and T calibration tokens; write H for the number of attention heads and d_{ff} for the MLP width. Per layer, Algorithm S3 costs:

(i) *Statistics.* One calibration forward pass (shared by *all* calibration-based methods, hence excluded from the timing comparisons). On top of it, the unstructured/diagonal statistics $\mathbf{G}_{jj} = \|\mathbf{X}_j\|_2^2$ cost $O(Ti)$, exactly as Wanda. The $N:M$ variant accumulates only the *block-diagonal* Grams $\{\mathbf{G}_{BB}\}$: i/M blocks of size $M \times M$, i.e. $O(TiM)$ time and $O(iM)$ memory—versus $O(Ti^2)$ time and $O(i^2)$ memory for the full Gram that SparseGPT builds and inverts. Structured statistics cost $O(Td_{\text{ff}})$ (neuron energies), $O(THd_h d_{\text{model}} + TH^2 d_{\text{model}})$ (head/group Gram; H is a few tens, so this is negligible), and the weight Grams $\mathbf{M} = \mathbf{V}^\top \mathbf{V}$ are activation-free one-off matrix products.

(ii) *Shrinkage and permutation.* The Ledoit–Wolf intensity ρ is a closed-form ratio of statistics already accumulated, $O(iM)$. The cheap (v2) permutation (Algorithm S1) sorts per-channel statistics and deals

channels round-robin, $O(i \log i)$ once per layer and shared across all rows; the spectral alternative (Algorithm S2) requires an eigendecomposition of an $i \times i$ affinity, $O(i^3)$ worst case, which is why the cheap variant is the one that scales to the 18,944-wide layers of Qwen2.5-7B (Table III).

(iii) *Selection.* Uncompensated $N:M$ selection evaluates, per row and block, $\binom{M}{N}$ candidate patterns, each a $(M-N) \times (M-N)$ quadratic form: $O(o \cdot \frac{i}{M} \cdot \binom{M}{N} (M-N)^2)$ total, i.e. $O(o i)$ up to the small constant $\binom{M}{N} (M-N)^2 / M$ (24 for 2:4, 140 for 4:8)—Wanda-class, and executed as a handful of batched tensor contractions (single-pass; Appendix A.5). Exact structured selection enumerates $\binom{H}{|P|}$ subsets of an $H \times H$ Gram only when that count is small; otherwise the diagonal ranking costs a sort and the greedy rule costs $O(|\text{units}|)$ per removal.

(iv) *Optional compensation.* Each block needs the $\binom{M}{N}$ Schur complements of $\mathbf{G}_{BB}^{-1}$, i.e. $O(\frac{i}{M} \binom{M}{N} M^3)$ shared across all o rows, plus an $O(o i M)$ weight update—still far below SparseGPT’s $O(i^3)$ layer inverse, and consistent with the measured $\sim 2\times$ overhead of “+comp.” in Table S15.

(v) *Memory.* The sweep is layer-sequential: one transformer block plus $O(iM)$ (or $O(H^2 + d_{\text{ff}})$ structured) statistics reside on the GPU at a time, so peak memory is dominated by the block weights themselves.

In total the selection overhead beyond the shared forward pass is linear in the number of weights with small constants, matching the wall-clock measurements of Table S15.

B Proofs

We first record a standard linear-algebra fact used by Propositions 2 and 3 and by the block-OBS derivation, and prove it for self-containedness.

B.1 A Schur-Complement Lemma

Lemma 1 (Schur complement: inverse block and PSD ordering). *Let $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{PP} & \mathbf{A}_{PK} \\ \mathbf{A}_{KP} & \mathbf{A}_{KK} \end{pmatrix} \succ 0$ be symmetric positive definite with $\mathbf{A}_{KP} = \mathbf{A}_{PK}^\top$, and let $\mathbf{S} = \mathbf{A}_{PP} - \mathbf{A}_{PK} \mathbf{A}_{KK}^{-1} \mathbf{A}_{KP}$ be the Schur complement of $\mathbf{A}_{KK}$. Then*

- (i) $\mathbf{S} \succ 0$ and $[\mathbf{A}^{-1}]_{PP} = \mathbf{S}^{-1}$, hence $([\mathbf{A}^{-1}]_{PP})^{-1} = \mathbf{S}$;
- (ii) $[\mathbf{A}^{-1}]_{KP} = -\mathbf{A}_{KK}^{-1} \mathbf{A}_{KP} \mathbf{S}^{-1}$;
- (iii) $\mathbf{S} \preceq \mathbf{A}_{PP}$ in the PSD order, with equality iff $\mathbf{A}_{PK} = \mathbf{0}$.

Proof. Write the block- $LDL^\top$ factorization

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{PK} \mathbf{A}_{KK}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{S} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{KK} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{KK}^{-1} \mathbf{A}_{KP} & \mathbf{I} \end{pmatrix},$$

which is verified by direct multiplication. The outer factors are unit triangular (determinant 1, hence invertible), so $\mathbf{A} \succ 0$ forces the middle block-diagonal factor to be positive definite; in particular $\mathbf{S} \succ 0$. Inverting the factorization (the triangular inverses are obtained by negating their off-diagonal blocks) gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{KK}^{-1} \mathbf{A}_{KP} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{S}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{KK}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{PK} \mathbf{A}_{KK}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix},$$

whose (P, P) block is $\mathbf{S}^{-1}$, proving (i), and whose (K, P) block is $-\mathbf{A}_{KK}^{-1} \mathbf{A}_{KP} \mathbf{S}^{-1}$, proving (ii). For (iii), $\mathbf{A}_{PP} - \mathbf{S} = \mathbf{A}_{PK} \mathbf{A}_{KK}^{-1} \mathbf{A}_{KP} = \mathbf{Z}^\top \mathbf{Z} \succeq 0$ with $\mathbf{Z} = \mathbf{A}_{KK}^{-1/2} \mathbf{A}_{KP}$ (the inverse square root exists as $\mathbf{A}_{KK} \succ 0$); thus $\mathbf{S} \preceq \mathbf{A}_{PP}$, and equality holds iff $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$, i.e. $\mathbf{Z} = \mathbf{0}$, i.e. $\mathbf{A}_{PK} = \mathbf{0}$. $\square$

B.2 Selection Guarantees (Proposition 1, Theorem 1)

Proof of Proposition 1. Fix an output row r with weights $\boldsymbol{\theta}_r \in \mathbb{R}^i$ and let P_r be the set of input indices pruned in that row. Write $\mathbf{m}_r \in \{0, 1\}^i$ for the keep-mask of row r , so the pruned row is $\mathbf{m}_r \odot \boldsymbol{\theta}_r$ with $(\mathbf{m}_r)_j = 0$ iff $j \in P_r$. The Frobenius objective (1) separates over rows because $\|\mathbf{A}\|_F^2 = \sum_r \|\mathbf{A}_{r,:}\|_2^2$; hence it suffices to analyze a single row.

The dense and pruned outputs of row r over the T calibration tokens are the T -vectors $\mathbf{y}_r = \boldsymbol{\theta}_r^\top \mathbf{X}$ and $\hat{\mathbf{y}}_r = (\mathbf{m}_r \odot \boldsymbol{\theta}_r)^\top \mathbf{X}$. Their difference keeps only the pruned coordinates,

$$\mathbf{y}_r - \hat{\mathbf{y}}_r = ((\mathbf{1} - \mathbf{m}_r) \odot \boldsymbol{\theta}_r)^\top \mathbf{X} = \sum_{j \in P_r} \theta_{rj} \mathbf{X}_{j,:} = \boldsymbol{\theta}_{r,P_r}^\top \mathbf{X}_{P_r},$$

where $\mathbf{X}_{P_r} \in \mathbb{R}^{|P_r| \times T}$ stacks the pruned channels' activation rows. Taking the squared norm and using $\mathbf{X}_{P_r} \mathbf{X}_{P_r}^\top = \mathbf{G}_{P_r, P_r}$ (the $P_r \times P_r$ submatrix of $\mathbf{G} = \mathbf{X} \mathbf{X}^\top$, by definition of $\mathbf{G}$),

$$\|\mathbf{y}_r - \hat{\mathbf{y}}_r\|_2^2 = (\boldsymbol{\theta}_{r,P_r}^\top \mathbf{X}_{P_r}) (\boldsymbol{\theta}_{r,P_r}^\top \mathbf{X}_{P_r})^\top = \boldsymbol{\theta}_{r,P_r}^\top \mathbf{X}_{P_r} \mathbf{X}_{P_r}^\top \boldsymbol{\theta}_{r,P_r} = \boldsymbol{\theta}_{r,P_r}^\top \mathbf{G}_{P_r, P_r} \boldsymbol{\theta}_{r,P_r}.$$

Summing over rows gives the stated exact error $\sum_r \boldsymbol{\theta}_{r,P_r}^\top \mathbf{G}_{P_r, P_r} \boldsymbol{\theta}_{r,P_r}$, which is finite and nonnegative since $\mathbf{G} \succeq 0$ by (A2).

For the diagonal claim, replacing $\mathbf{G}_{P_r, P_r}$ by its diagonal $\text{diag}(\mathbf{G}_{P_r, P_r})$ drops every cross term $j \neq k$ and leaves

$$\sum_{j \in P_r} \theta_{rj}^2 \mathbf{G}_{jj} = \sum_{j \in P_r} \theta_{rj}^2 \|\mathbf{X}_j\|_2^2 = \sum_{j \in P_r} (|\theta_{rj}| \|\mathbf{X}_j\|_2)^2 = \sum_{j \in P_r} \sigma_{rj}^2,$$

using $\mathbf{G}_{jj} = \|\mathbf{X}_j\|_2^2$. At a fixed cardinality $|P_r| = k$, the diagonal surrogate $\sum_{j \in P_r} \sigma_{rj}^2$ is a sum of k nonnegative terms, so it is minimized by choosing the k indices with the smallest values of σ_{rj}^2 ; since $\sigma_{rj} \geq 0$ and $t \mapsto t^2$ is strictly increasing on $[0, \infty)$, the k smallest σ_{rj}^2 are the k smallest σ_{rj} . Selecting these as the pruned set is exactly the Wanda rule. (Note this uses that the objective is *separable* across indices; it is *not* the same as minimizing the *sum* $\sum_{j \in P_r} \sigma_{rj}$, which is not separable—the equivalence holds specifically because we choose the k smallest entries of a sum of independent nonnegative terms.) Finally, if $\mathbf{G}_{P_r, P_r}$ is diagonal for every candidate P_r , the cross terms vanish identically, so the diagonal surrogate equals the exact error (5) and Wanda's selection attains the minimum of the exact objective. $\square$

Proof of Theorem 1. All sets below range over a fixed candidate family $\mathcal{F}$ of pruned sets of a common target cardinality (e.g. all $P \subseteq B$ with $|P| = M - N$ within a block, or all index sets of a fixed size in a row); $P_g, P^* \in \mathcal{F}$, and the common cardinality is what makes the comparison $\text{Diag}(P_g) \leq \text{Diag}(P^*)$ meaningful. By (5) and bilinearity of the quadratic form,

$$\text{Sal}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{P,P} \boldsymbol{\theta}_{r,P} = \underbrace{\sum_{j \in P} \theta_{rj}^2 \mathbf{G}_{jj}}_{\text{Diag}(P)} + \underbrace{\sum_{j \neq k \in P} \theta_{rj} \theta_{rk} \mathbf{G}_{jk}}_{E(P)},$$

where the $j \neq k$ sum is over ordered pairs. By the triangle inequality applied termwise, $|E(P)| \leq \sum_{j \neq k \in P} |\theta_{rj} \theta_{rk}| |\mathbf{G}_{jk}| = \text{Off}(P)$, whence

$$\text{Diag}(P) - \text{Off}(P) \leq \text{Sal}(P) \leq \text{Diag}(P) + \text{Off}(P) \quad \text{for all } P \in \mathcal{F}. \quad (*)$$

Now bound the two selections. Using the upper half of $(*)$ at P_g and then $\text{Diag}(P_g) \leq \text{Diag}(P^*)$ (optimality of P_g for Diag over $\mathcal{F}$),

$$\text{Sal}(P_g) \leq \text{Diag}(P_g) + \text{Off}(P_g) \leq \text{Diag}(P^*) + \text{Off}(P_g),$$

and using the lower half of $(*)$ at P^* , $\text{Sal}(P^*) \geq \text{Diag}(P^*) - \text{Off}(P^*)$. Subtracting, the common $\text{Diag}(P^*)$ cancels:

$$\text{Sal}(P_g) - \text{Sal}(P^*) \leq \text{Off}(P_g) + \text{Off}(P^*) \leq 2 \max_{P \in \mathcal{F}} \text{Off}(P).$$

The left-hand gap is nonnegative because P^* minimizes Sal over $\mathcal{F}$. Finally, if $\mathbf{G}$ restricted to the candidate support is diagonal then $\mathbf{G}_{jk} = 0$ for all $j \neq k$ appearing in any $P \in \mathcal{F}$, so $\text{Off} \equiv 0$, the sandwich $(*)$ collapses to $\text{Sal} \equiv \text{Diag}$ on $\mathcal{F}$, and hence $\text{Sal}(P_g) = \text{Diag}(P_g) = \min_{\mathcal{F}} \text{Diag} = \min_{\mathcal{F}} \text{Sal} = \text{Sal}(P^*)$; that is, P_g attains the exact optimum *in value* (the optimal sets may still differ under ties). $\square$

B.3 Compensation Guarantees (Propositions 2 and 3)

Proof of Proposition 2. By (A2) take $\mathbf{G}_{BB} \succ 0$ (else use $\mathbf{G}_{BB} + \lambda \mathbf{I}$), so $\mathbf{G}_{KK} \succ 0$ as a principal submatrix and Lemma 1 applies with $\mathbf{A} = \mathbf{G}_{BB}$. By parts (i) and (iii) of the lemma,

$$([\mathbf{G}_{BB}^{-1}]_{PP})^{-1} = \mathbf{S}_P = \mathbf{G}_{PP} - \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \preceq \mathbf{G}_{PP},$$

with equality iff $\mathbf{G}_{PK} = \mathbf{0}$. Evaluating both quadratic forms at the common vector $\boldsymbol{\theta}_{r,P}$ gives $\text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{S}_P \boldsymbol{\theta}_{r,P} \leq \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P} = \text{Sal}(P)$, so block-local compensation never increases the block reconstruction error and coincides with the uncompensated cost (7) exactly when the pruned and kept channels are $\mathbf{G}$ -orthogonal. That $\text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{S}_P \boldsymbol{\theta}_{r,P}$ is the optimal value of the OBS sub-problem, and the corresponding update in (9), are derived in Appendix A.3.3 (using Lemma 1(ii)). $\square$

Proposition 1 (Reconstruction error of GRES, with and without compensation; full form of main-text Proposition 3). *Fix a row r and a block B with pruned set P , kept set $K = B \setminus P$, and let $\lambda_{\min}(\cdot), \lambda_{\max}(\cdot)$ denote extreme eigenvalues. Then*

$$\begin{aligned} (\text{no comp.}) \quad \text{Sal}(P) &= \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P}, \\ \lambda_{\min}(\mathbf{G}_{PP}) \|\boldsymbol{\theta}_{r,P}\|_2^2 &\leq \text{Sal}(P) \leq \lambda_{\max}(\mathbf{G}_{PP}) \|\boldsymbol{\theta}_{r,P}\|_2^2, \end{aligned} \quad (3)$$

$$(\text{comp.}) \quad \text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{S}_P \boldsymbol{\theta}_{r,P}, \quad 0 \leq \text{Sal}_{\text{comp}}(P) \leq \text{Sal}(P), \quad (4)$$

where $\mathbf{S}_P = \mathbf{G}_{PP} - \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \succeq 0$ is the Schur complement; the compensation gain is exactly

$$\text{Sal}(P) - \text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P} = \|\mathbf{G}_{KK}^{-1/2} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P}\|_2^2 \geq 0; \quad (5)$$

and the diagonal (Wanda) surrogate brackets the no-compensation error as

$$|\text{Sal}(P) - \text{Diag}(P)| \leq \text{Off}(P) \leq (|P| - 1) \rho_B \text{Diag}(P), \quad \rho_B := \max_{j \neq k \in B} \frac{|\mathbf{G}_{jk}|}{\sqrt{\mathbf{G}_{jj} \mathbf{G}_{kk}}} \in [0, 1], \quad (6)$$

with $\text{Diag}(P) = \sum_{j \in P} \sigma_{rj}^2$. Summing over rows and blocks, the total layer reconstruction error of each variant is the sum of the corresponding block terms.

Proof of Proposition 1 (hence of main-text Proposition 3). (3): $\text{Sal}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P}$ is a Rayleigh quotient of the symmetric PSD matrix $\mathbf{G}_{PP}$ scaled by $\|\boldsymbol{\theta}_{r,P}\|_2^2$, so it lies between $\lambda_{\min}(\mathbf{G}_{PP}) \|\boldsymbol{\theta}_{r,P}\|_2^2$ and $\lambda_{\max}(\mathbf{G}_{PP}) \|\boldsymbol{\theta}_{r,P}\|_2^2$; nonnegativity is $\mathbf{G}_{PP} \succeq 0$.

(4) and (5): by Proposition 2 and Lemma 1, $\text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top \mathbf{S}_P \boldsymbol{\theta}_{r,P}$ with the Schur complement $\mathbf{S}_P = \mathbf{G}_{PP} - \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \succeq 0$. Subtracting,

$$\text{Sal}(P) - \text{Sal}_{\text{comp}}(P) = \boldsymbol{\theta}_{r,P}^\top (\mathbf{G}_{PP} - \mathbf{S}_P) \boldsymbol{\theta}_{r,P} = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PK} \mathbf{G}_{KK}^{-1} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P} = \|\mathbf{G}_{KK}^{-1/2} \mathbf{G}_{KP} \boldsymbol{\theta}_{r,P}\|_2^2 \geq 0,$$

using $\mathbf{G}_{KK} \succ 0$ (A2) to define $\mathbf{G}_{KK}^{-1/2}$. Hence $0 \leq \text{Sal}_{\text{comp}}(P) \leq \text{Sal}(P)$, the lower bound following from $\mathbf{S}_P \succeq 0$ (Lemma 1(i)).

(6): the first inequality is Theorem 1. For the second, bound each off-diagonal term by Cauchy–Schwarz in the activation inner product, $|\mathbf{G}_{jk}| \leq \rho_B \sqrt{\mathbf{G}_{jj} \mathbf{G}_{kk}}$ for $j \neq k \in B$ by definition of ρ_B , and $|\theta_{rj} \theta_{rk}| \sqrt{\mathbf{G}_{jj} \mathbf{G}_{kk}} = \sigma_{rj} \sigma_{rk}$. Thus

$$\text{Off}(P) = \sum_{j \neq k \in P} |\theta_{rj} \theta_{rk}| |\mathbf{G}_{jk}| \leq \rho_B \sum_{j \neq k \in P} \sigma_{rj} \sigma_{rk} = \rho_B \left[\left(\sum_{j \in P} \sigma_{rj} \right)^2 - \sum_{j \in P} \sigma_{rj}^2 \right].$$

By Cauchy–Schwarz $(\sum_{j \in P} \sigma_{rj})^2 \leq |P| \sum_{j \in P} \sigma_{rj}^2$, so the bracket is at most $(|P| - 1) \sum_{j \in P} \sigma_{rj}^2 = (|P| - 1) \text{Diag}(P)$, giving $\text{Off}(P) \leq (|P| - 1) \rho_B \text{Diag}(P)$. Finally $\rho_B \in [0, 1]$ is a correlation coefficient (Cauchy–Schwarz on $\langle \mathbf{X}_j, \mathbf{X}_k \rangle$). The layer-total claim follows by summing the exact per-block, per-row identities, which are independent across rows and blocks. $\square$

B.4 Granularity Reduction

Proof of the granularity-reduction claim in main-text Remark 1. Specialize the general saliency (6) to single-weight units in row r , whose removal directions are the scalars $\delta_j(t) = \theta_{rj} \mathbf{X}_{jt}$. Then for all j, k ,

$$\Gamma_{jk} = \sum_{t=1}^T \delta_j(t) \delta_k(t) = \theta_{rj} \theta_{rk} \sum_{t=1}^T \mathbf{X}_{jt} \mathbf{X}_{kt} = \theta_{rj} \theta_{rk} \mathbf{G}_{jk} = (\boldsymbol{\theta}_r \boldsymbol{\theta}_r^\top \odot \mathbf{G})_{jk},$$

so $\boldsymbol{\Gamma} = \boldsymbol{\theta}_r \boldsymbol{\theta}_r^\top \odot \mathbf{G}$ and $\text{Sal}(P) = \mathbf{1}_P^\top \boldsymbol{\Gamma}_{PP} \mathbf{1}_P = \boldsymbol{\theta}_{r,P}^\top \mathbf{G}_{PP} \boldsymbol{\theta}_{r,P}$, which is (5). Approximating $\boldsymbol{\Gamma}$ by its diagonal keeps only $\Gamma_{jj} = \theta_{rj}^2 \mathbf{G}_{jj} = \sigma_{rj}^2$, recovering the squared Wanda objective of Proposition 1. Thus the diagonal, single-weight instance of (6) is exactly Wanda, and the $N:M$ and structured criteria are obtained from the *same* $\boldsymbol{\Gamma}$ by enlarging the unit and retaining its off-diagonal block. $\square$

C Additional Experimental Results and Reproducibility

C.1 Semi-Structured Pruning: Per-Task Accuracy

Table S4 reports the per-task accuracies underlying the LLaMA-2-13B zero-shot means in main-text Table I. Magnitude trails the calibration-based methods on every task (the usual outlier-channel effect [2]); Wanda, SparseGPT, and RIA are within ~ 0.7 points at unstructured 50%, while at the restrictive 2:4 pattern SparseGPT’s weight reconstruction opens a ~ 2 -point lead. GRES coincides with Wanda at unstructured 50% by Proposition 1; its semi-structured means (59.06 at 4:8, 54.72 at 2:4, the best forward-only results) appear in main-text Table I, with the per-task breakdown in Table S5.

Table S4: Per-task zero-shot accuracy ($\uparrow$, %) on LLaMA-2-13B underlying the means in main-text Table I. Tasks: BoolQ, RTE, HellaSwag (HS), WinoGrande (WG), ARC-easy (ARC-e), ARC-challenge (ARC-c), OpenBookQA (OBQA). GRES coincides with Wanda at unstructured 50% (Proposition 1).

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>Unstructured 50%</i>								
Magnitude	57.65	55.96	54.40	65.19	70.54	38.40	27.20	52.76
Wanda	81.04	60.65	57.20	71.27	76.47	43.43	31.20	60.18
SparseGPT	81.19	61.73	55.75	70.01	74.28	43.17	31.00	59.59
RIA	80.80	58.48	56.77	71.11	76.09	41.04	32.20	59.50
<i>Semi-structured 4:8</i>								
Magnitude	63.24	57.76	54.02	64.48	68.43	35.84	26.00	52.83
Wanda	79.91	63.90	52.25	69.14	73.95	39.68	30.00	58.40
SparseGPT	80.70	60.29	52.37	71.35	74.03	40.78	28.80	58.33
RIA	79.94	59.21	52.85	68.35	73.65	38.48	29.00	57.35
<i>Semi-structured 2:4</i>								
Magnitude	65.66	53.79	50.12	62.04	62.29	31.83	23.40	49.87
Wanda	76.06	55.96	46.42	67.09	68.01	34.90	25.40	53.40
SparseGPT	80.76	61.37	48.16	68.59	69.40	37.12	27.20	56.09
RIA	76.79	56.68	47.62	67.25	69.70	34.90	26.40	54.19

C.2 Structured Pruning: LLaMA-2-13B and Block-Local Compensation

Table S6 reports the LLaMA-2-13B structured-pruning results referenced in the main text (Section IV-C). The same ordering observed at 7B carries over: GRES-head/group give the best perplexity at both 20% and 50% at roughly $9\times$ less pruning-stage time than SliceGPT.

Table S5: Per-task zero-shot accuracy ($\uparrow$, %) of single-pass GRES across six models and three sparsity patterns, under the same 0-shot harness as all other tables (the unstructured rows use the diagonal metric and hence track Wanda; the $N:M$ rows use the cheap-permutation single-pass selection).

Model	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>Unstructured 50%</i>								
LLaMA-2-7B	76.36	53.43	52.87	67.88	72.64	38.99	31.00	56.17
LLaMA-2-13B	81.04	60.65	57.20	71.27	76.47	43.43	31.20	60.18
LLaMA-3.2-3B	68.75	54.51	45.35	63.93	65.24	31.91	24.80	50.64
Mistral-7B	83.73	57.76	55.11	70.56	75.29	43.34	26.60	58.91
Qwen2.5-7B	83.76	77.26	53.23	71.11	76.39	42.06	30.60	62.06
Qwen2.5-3B	71.13	75.09	45.67	65.82	71.63	37.37	26.00	56.10
<i>Semi-structured 4:8</i>								
LLaMA-2-7B	68.65	54.51	48.64	66.61	69.70	35.75	28.40	53.18
LLaMA-2-13B	80.18	67.51	52.96	70.72	71.89	39.76	30.40	59.06
LLaMA-3.2-3B	66.42	53.43	39.42	60.54	59.26	28.24	22.20	47.07
Mistral-7B	80.49	56.32	49.67	68.98	70.54	36.35	25.00	55.34
Qwen2.5-7B	82.23	76.90	49.57	69.61	75.67	42.41	28.20	60.66
Qwen2.5-3B	62.51	70.40	40.88	64.09	66.37	32.08	25.20	51.65
<i>Semi-structured 2:4</i>								
LLaMA-2-7B	69.88	54.87	43.07	64.40	64.06	29.78	25.20	50.18
LLaMA-2-13B	77.28	57.40	48.19	67.17	69.53	35.49	28.00	54.72
LLaMA-3.2-3B	63.52	52.71	34.56	56.20	52.15	24.49	17.20	42.97
Mistral-7B	68.93	57.40	44.00	66.06	66.33	30.20	19.60	50.36
Qwen2.5-7B	78.69	73.65	45.18	67.17	71.09	38.31	23.60	56.81
Qwen2.5-3B	63.03	49.46	36.51	57.70	59.47	27.56	19.60	44.76

Table S6: LLaMA-2-13B structured pruning: WikiText-2 perplexity ($\downarrow$) at 20%/50% and pruning time at 20%. Best non-dense per column in **bold**.

Method	20%	50%	Time (20%)
Dense	4.88	4.88	—
SliceGPT	6.00	17.15	1668 s
2SSP	7.10	23.60	434 s
SLEB	6.80	31.18	—
GRES-head	5.85	13.53	178 s
GRES-neuron	6.07	14.69	164 s
GRES-group	5.85	13.53	178 s

Effect of the optional block-local compensation (“+C”). Table S7 isolates the optional block-local OBS compensation (9) on the KV-group variant, comparing GRES-group against GRES-group+C at 20% and 50%. By Proposition 2, compensation never increases the *on-calibration* block reconstruction error; downstream, however, its effect on perplexity is mixed. It yields a clear gain only on LLaMA-2-7B at 50% (11.74 $\rightarrow$ 11.14), and is otherwise neutral or marginally worse—within run-to-run noise (e.g. 33.58 $\rightarrow$ 33.61 on LLaMA-3.2-3B and 16.75 $\rightarrow$ 16.77 on Mistral-7B at 50%). The gain is realized only when within-group correlations are strong and the calibration Gram is well-estimated, so that the surviving channels can genuinely absorb part of the pruning error; elsewhere the small reconstruction improvement does not translate into lower perplexity and is swamped by estimation noise. Because “+C” also requires a per-block $M \times M$ Schur inverse (which dominates the cost at 13B; cf. Table S15), we disable it by default and report it as an optional tier. This mirrors the $N:M$ setting (main-text Table III), where compensation is likewise mixed, helping slightly on three of the six models: the value of compensation depends jointly on the granularity and the model, and by Proposition 2 it is never harmful to the calibration objective in either case, exactly as the theory guarantees.

Table S7: Effect of the optional block-local OBS compensation (“+C”) on the KV-group variant: WikiText-2 perplexity ($\downarrow$) at 20%/50% structured pruning. Lower per pair in **bold** when the difference exceeds run-to-run noise. Compensation helps clearly only on LLaMA-2-7B at 50%; elsewhere it is neutral or within noise, consistent with Proposition 2 (it never increases the on-calibration block reconstruction error) while adding a per-block $M \times M$ inverse. $\text{int}(4 \times 0.2) = 0$ KV groups removed at 20% (no pruning), so the pair is not directly comparable there.

Model	20%		50%	
	group	+C	group	+C
LLaMA-2-7B	6.63	6.63	11.74	11.14
LLaMA-3.2-3B	8.75	8.76	33.58	33.61
Mistral-7B	6.21	6.21	16.75	16.77
Qwen2.5-7B	6.85*	8.62*	55.98	56.28

C.3 Structured Pruning: FLAP Comparison and Per-Task Accuracy

FLAP [9] is omitted from main-text Table II because its modeling code supports only non-GQA (MHA) attention, so it runs on the LLaMA-2 models but not on the GQA models (LLaMA-3.2-3B, Mistral-7B, Qwen2.5-7B/3B). For completeness, Table S8 reports its structured results on LLaMA-2-7B/13B under the same protocol; GRES beats it on both metrics (e.g. 57.4 vs. 51.1 accuracy at 20% on LLaMA-2-7B). FLAP’s per-task accuracies appear in Tables S9–S10.

Table S8: FLAP [9] structured pruning on the MHA LLaMA-2 models: WikiText-2 perplexity (PPL $\downarrow$) and mean zero-shot accuracy (Acc $\uparrow$, %) at 20%/50%. FLAP supports only non-GQA attention and is therefore excluded from the main table and the GQA models; “–” not measured.

Model	20%		50%	
	PPL	Acc	PPL	Acc
LLaMA-2-7B	16.33	51.07	43.07	37.33
LLaMA-2-13B	–	54.21	–	46.86

Per-task accuracy: structured pruning. Tables S9–S14 give the per-task zero-shot accuracies underlying the structured-pruning means in main-text Table II, for all methods at 20% and 50%. All values are 0-shot **acc** (not **acc_norm**) from the single shared harness, so they are directly comparable across methods. “–” marks FLAP on GQA models (unsupported); “N/A $\uparrow$ ” marks GRES-group on Qwen2.5-3B (only two KV groups); “*” marks the no-pruning case ($\text{int}(4 \times 0.2) = 0$ KV groups on Qwen2.5-7B, hence equal to dense). On the MHA models (LLaMA-2-7B/13B) GRES-group $\equiv$ GRES-head by construction. For each ratio, the best (highest) value in every column—each task and the mean—is in **bold**; the no-pruning * row is excluded from the comparison.

C.4 Pruning Efficiency: Full Timing Data

Table S15 reports the wall-clock 2:4 pruning times summarized in main-text Section IV-D, and Fig. S1 plots the quality–cost trade-off on all six models (main-text Fig. 4 shows the LLaMA-2 panels). Two notes on interpreting the margins. The order-of-magnitude gap to SparseGPT is *algorithmic*: SparseGPT performs a layer-wise Hessian inverse and a weight-update pass that GRES avoids entirely. The margin over Wanda and RIA, by contrast, partly reflects implementation: GRES’s $N:M$ selection is fully vectorized over blocks and rows, whereas the reference Wanda/RIA implementations select each block with a Python loop; all methods are timed as released, on identical inputs, with GPU synchronization (cf. the complexity analysis of Appendix A.6).

Table S9: Per-task zero-shot accuracy ($\uparrow$, %) on **LLaMA-2-7B** under 20%/50% structured pruning.

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	71.10	57.76	50.49	66.85	67.72	35.92	28.20	54.01
2SSP	65.93	48.74	49.84	63.14	68.43	34.13	25.40	50.80
SLEB	65.26	59.57	46.82	59.43	64.90	32.76	25.80	50.65
FLAP	57.25	58.84	48.10	63.61	67.26	34.22	28.20	51.07
GR _E S-head	74.13	54.15	56.34	68.27	75.13	42.83	30.80	57.38
GR _E S-neuron	69.82	60.29	55.45	68.19	71.68	39.25	29.40	56.30
GR _E S-group	74.13	54.15	56.34	68.27	75.13	42.83	30.80	57.38
<i>50% pruned</i>								
SliceGPT	62.29	52.71	32.56	57.85	34.13	20.73	13.80	39.15
2SSP	51.25	50.54	35.09	51.85	42.55	22.95	19.20	39.06
SLEB	43.94	52.71	29.42	50.28	36.49	22.44	15.20	35.78
FLAP	38.38	54.51	33.16	54.30	37.88	23.29	19.80	37.33
GR _E S-head	62.66	52.35	41.04	52.25	63.01	29.27	23.60	46.31
GR _E S-neuron	64.40	56.32	41.60	62.83	47.81	26.88	19.20	45.58
GR _E S-group	62.66	52.35	41.04	52.25	63.01	29.27	23.60	46.31

Table S10: Per-task zero-shot accuracy ($\uparrow$, %) on **LLaMA-2-13B** under 20%/50% structured pruning.

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	78.75	60.65	54.47	70.17	71.80	39.93	30.60	58.05
2SSP	72.29	59.57	55.17	67.48	73.40	40.70	30.00	56.94
SLEB	67.19	58.12	51.42	63.46	70.92	38.99	29.80	54.27
FLAP	67.16	54.87	51.94	67.17	69.44	37.88	31.00	54.21
GR _E S-head	82.75	64.26	60.43	72.38	79.50	47.10	35.60	63.15
GR _E S-neuron	79.24	61.37	58.95	72.69	74.66	44.20	33.40	60.64
GR _E S-group	82.75	64.26	60.43	72.38	79.50	47.10	35.60	63.15
<i>50% pruned</i>								
SliceGPT	62.26	52.71	34.38	60.54	36.74	22.95	15.40	40.71
2SSP	64.56	55.23	39.24	58.48	48.44	26.96	19.20	44.59
SLEB	61.83	49.10	35.31	54.62	41.71	24.57	18.00	40.73
FLAP	64.43	58.48	37.59	58.72	54.55	28.84	25.40	46.86
GR _E S-head	64.95	48.01	41.88	55.09	39.98	21.08	22.00	41.86
GR _E S-neuron	70.43	52.35	46.24	67.17	53.62	30.72	23.40	49.13
GR _E S-group	64.95	48.01	41.88	55.09	39.98	21.08	22.00	41.86

C.5 Component Ablation at 4:8

Table S16 repeats the component ablation of main-text Table III under the milder 4:8 pattern, and the ordering of the components is the same. The looser constraint leaves fewer forced joint decisions per block, so the off-diagonal criterion and Gram shrinkage have correspondingly less coupling to exploit and their effect is small; the channel permutation provides the gain on every model, and the optional block-OBS compensation is essentially neutral on top of it. This is the behavior Theorem 1 predicts: the value of correlation-aware selection scales with the within-block correlation that the constraint forces the selector to confront, which is why the 2:4 gains of main-text Table III are the larger ones.

When block-local compensation is worthwhile. Proposition 2 guarantees that the update never increases the *block reconstruction error*, and Proposition 3 identifies the size of the gain exactly: it is the energy of the pruned weights projected through $\mathbf{G}_{BB}$ onto the kept channels. The gain is therefore largest when the pruned and kept channels of a block remain strongly $\mathbf{G}$ -coupled, and vanishes when they are $\mathbf{G}$ -orthogonal. This is what the measurements show. Once the channel permutation has already redistributed importance across blocks, the residual coupling each block must absorb is small, so the additional reconstruction gain

Table S11: Per-task zero-shot accuracy ($\uparrow$, %) on **LLaMA-3.2-3B** under 20%/50% structured pruning.

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	70.98	54.51	45.99	65.82	59.85	29.86	22.20	49.89
2SSP	57.74	55.96	43.40	61.17	60.90	29.61	24.00	47.54
SLEB	53.55	51.99	39.28	53.99	56.69	26.11	20.20	43.11
FLAP	—	—	—	—	—	—	—	—
GRES-head	66.45	49.46	52.16	60.62	72.10	38.14	26.40	52.19
GRES-neuron	68.50	48.38	47.31	64.33	63.47	32.34	27.20	50.22
GRES-group	70.18	62.45	53.88	63.93	73.48	41.55	30.60	56.58
<i>50% pruned</i>								
SliceGPT	62.14	52.71	29.26	51.38	31.82	17.66	14.00	37.00
2SSP	55.96	53.07	30.96	52.25	37.12	19.71	16.20	37.90
SLEB	38.17	51.62	27.36	52.64	31.27	19.28	15.00	33.62
FLAP	—	—	—	—	—	—	—	—
GRES-head	58.72	52.35	27.20	51.14	28.24	18.34	9.80	35.11
GRES-neuron	42.45	54.15	34.68	57.70	39.44	20.90	15.60	37.85
GRES-group	62.08	54.51	30.93	50.51	43.14	19.20	15.00	39.34

Table S12: Per-task zero-shot accuracy ($\uparrow$, %) on **Mistral-7B** under 20%/50% structured pruning.

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	79.02	59.93	51.67	69.85	68.43	35.41	24.60	55.56
2SSP	75.69	54.51	53.66	67.72	66.75	36.18	25.80	54.33
SLEB	65.87	52.71	48.25	62.19	68.60	35.24	25.80	51.24
FLAP	—	—	—	—	—	—	—	—
GRES-head	81.10	64.98	57.71	66.46	78.75	44.03	32.20	60.75
GRES-neuron	78.01	55.60	56.98	71.19	71.93	39.16	28.00	57.27
GRES-group	82.23	62.82	59.01	72.53	79.88	48.46	32.20	62.45
<i>50% pruned</i>								
SliceGPT	62.14	52.71	30.70	54.22	32.37	18.09	11.80	37.43
2SSP	63.15	52.71	34.91	55.72	37.29	19.97	16.40	40.02
SLEB	38.17	53.79	31.05	50.36	32.95	24.06	16.60	35.28
FLAP	—	—	—	—	—	—	—	—
GRES-head	59.48	53.07	34.71	52.49	54.50	20.73	17.80	41.83
GRES-neuron	68.84	52.71	37.12	60.22	42.55	24.23	16.40	43.15
GRES-group	62.45	52.71	31.50	52.64	47.14	19.03	15.00	40.07

is small—and because end-task perplexity is a different quantity from layer reconstruction error, a small reconstruction gain need not move perplexity in a consistent direction (Tables S16 and S7). Compensation is accordingly most useful where the block error itself is the quantity of interest, where the mask is fixed externally, or where the permutation is unavailable—for instance when the channel order must be preserved for a downstream kernel. We leave it off by default and report it as an option, at roughly $2\times$ the pruning-stage time (Table S15).

C.6 Composability of the Off-Diagonal Lift

Table S17 pairs GRES’s block-Gram off-diagonal selection with two diagonal base scores, complementing the component ablation of main-text Table III.

C.7 Reproducibility

Code and scripts. Our implementation is a self-contained Python package with *no custom CUDA kernels*. A single entry point `main.py` supports all three granularities (unstructured, $N:M$, structured) through

Table S13: Per-task zero-shot accuracy ($\uparrow$, %) on **Qwen2.5-7B** under 20%/50% structured pruning. *GRES-group at 20% removes 0 KV groups (no pruning), so it equals dense.

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	79.91	68.23	49.11	66.30	73.65	38.48	27.40	57.58
2SSP	81.28	62.82	54.70	66.93	68.14	35.24	27.80	56.70
SLEB	51.41	51.99	46.86	56.59	72.98	36.35	30.60	49.54
FLAP	—	—	—	—	—	—	—	—
GRES-head	83.70	73.65	57.80	68.27	77.99	45.65	34.40	63.06
GRES-neuron	82.29	71.48	56.01	68.67	69.53	38.82	29.80	59.52
GRES-group	84.07	71.84	60.03	72.85	81.36	48.04	32.60	65.81*
<i>50% pruned</i>								
SliceGPT	61.96	52.71	30.27	54.46	36.78	17.15	14.00	38.19
2SSP	63.39	54.15	37.37	57.93	42.97	20.31	17.20	41.90
SLEB	46.51	53.07	30.94	50.20	42.80	19.11	17.60	37.18
FLAP	—	—	—	—	—	—	—	—
GRES-head	60.40	53.43	38.83	52.01	58.84	29.18	22.40	45.01
GRES-neuron	66.88	53.79	41.67	59.59	49.62	26.54	21.20	45.61
GRES-group	39.82	47.29	28.34	51.38	43.60	19.11	15.80	35.05

command-line arguments, and per-model scripts (`scripts/llama2_7b.sh`, etc.) reproduce all tables with one command per model. Code and scripts are available at <https://github.com/futuresun912/GRES>.

Environment.

Component	Version
GPU	NVIDIA RTX PRO 6000 Blackwell Max-Q (96 GB GDDR7, compute 12.0)
CPU / RAM	AMD EPYC 9455P / 502 GiB
PyTorch	2.11.0+cu130
Transformers	5.4.0
Python	3.13

Dependencies install with a single command and require no additional libraries:

```
pip install torch transformers datasets numpy accelerate
```

Model checkpoints. All six models are the public base (non-instruction-tuned) checkpoints, loaded in float16 through transformers:

Model	HuggingFace identifier
LLaMA-2-7B	<code>meta-llama/Llama-2-7b-hf</code>
LLaMA-2-13B	<code>meta-llama/Llama-2-13b-hf</code>
LLaMA-3.2-3B	<code>meta-llama/Llama-3.2-3B</code>
Mistral-7B	<code>mistralai/Mistral-7B-v0.1</code>
Qwen2.5-7B	<code>Qwen/Qwen2.5-7B</code>
Qwen2.5-3B	<code>Qwen/Qwen2.5-3B</code>

The exact checkpoint revisions are pinned in the released driver scripts, so a single command per model reproduces the reported numbers on the same weights.

Calibration and evaluation. Calibration uses 128 sequences sampled from C4 (streaming) at sequence length 2048 with seed 0; all methods (Wanda, SparseGPT, GRES) use *identical* calibration data for a fair comparison. We report WikiText-2 test-set perplexity (sliding window, sequence length 2048) and mean zero-shot accuracy over seven tasks—BoolQ, RTE, HellaSwag, WinoGrande, ARC-easy, ARC-challenge, and OpenBookQA—using the LM-Evaluation-Harness in the 0-shot setting with its default per-task metric (`acc`

Table S14: Per-task zero-shot accuracy ($\uparrow$, %) on **Qwen2.5-3B** under 20%/50% structured pruning. † GRES-group is N/A (only two KV groups).

Method	BoolQ	RTE	HS	WG	ARC-e	ARC-c	OBQA	Mean
<i>20% pruned</i>								
SliceGPT	69.63	55.60	42.48	62.27	63.51	29.78	24.60	49.70
2SSP	69.51	68.23	48.05	64.01	62.50	32.59	26.60	53.07
SLEB	62.26	54.51	45.00	54.54	71.04	34.73	27.00	49.87
FLAP	—	—	—	—	—	—	—	—
GRoS-head	72.97	77.62	52.17	60.46	76.01	43.60	29.00	58.83
GRoS-neuron	70.40	52.71	48.13	65.82	65.07	33.45	30.20	52.25
GRoS-group	N/A [†]							
<i>50% pruned</i>								
SliceGPT	62.32	52.71	28.54	52.17	34.76	17.75	11.20	37.06
2SSP	62.17	49.82	33.79	55.56	41.20	19.71	18.40	40.09
SLEB	62.11	51.62	29.55	50.67	43.10	20.31	19.00	39.48
FLAP	—	—	—	—	—	—	—	—
GRoS-head	47.09	52.71	26.89	50.51	39.86	17.41	14.00	35.50
GRoS-neuron	62.78	49.82	35.44	54.22	45.83	22.87	19.80	41.54
GRoS-group	N/A [†]							

Table S15: Pruning time (s, $\downarrow$) for 2:4 sparsity (setting of main-text Section IV-A; shared calibration forward excluded, GPU-synchronized). With single-pass, vectorized selection GRES has the lowest pruning-stage cost of the forward-only methods on every model; the margin over Wanda/RIA reflects vectorized versus loop-based block selection rather than an algorithmic difference, whereas the 40–85 $\times$ margin over SparseGPT is algorithmic (global Hessian inverse and weight update). “+comp.” roughly doubles GRES’s time yet stays below all baselines. Lowest per column in **bold**.

Method	LLaMA-2-7B	LLaMA-2-13B	LLaMA-3.2-3B	Mistral-7B	Qwen2.5-7B	Qwen2.5-3B
SparseGPT	90.6	146.3	59.6	102.2	100.8	68.3
Wanda	8.1	12.7	4.5	8.5	8.0	5.1
RIA	8.3	14.0	4.8	8.8	8.1	5.2
GRES (single-pass, no comp.)	1.8	3.6	0.8	2.0	1.9	0.8
GRES (+comp.)	3.9	7.8	1.8	4.3	4.1	1.9

for BoolQ, RTE, WinoGrande and ARC-easy; `acc_norm` for HellaSwag, ARC-challenge and OpenBookQA). The harness version and the evaluation batch size are pinned in the released driver scripts together with the package versions listed above. All reported numbers are point estimates under this single fixed protocol.

Choice of the calibration set. The calibration set enters GRES only through the Gram statistics, and its size interacts with the shrinkage intensity of Eq. (3) in a self-regulating way. With few sequences the off-diagonal estimates carry more sampling noise, the Ledoit–Wolf estimator returns a larger ρ , and the criterion falls back toward the diagonal rule, which needs no off-diagonal estimate at all; as the calibration set grows relative to the layer width, ρ decreases and more of the measured correlation is used. GRES therefore has no calibration-size hyperparameter to tune, and we keep the standard 128×2048 -token protocol of Wanda and SparseGPT for every method so that the comparison with prior work is direct. A systematic study of calibration size, source, and seed—together with the resulting ρ profile across layers—is a natural complement to this work.

Reproducing key results. Every main-text table reproduces from the released repository with a single command: the entry point `main.py` selects the granularity (unstructured, $N:M$, or structured), the sparsity ratio, and the correlation-aware options through command-line flags, and per-model driver scripts run the full pipeline—calibration, pruning, perplexity, and zero-shot evaluation—end to end, with the expected runtimes listed above.

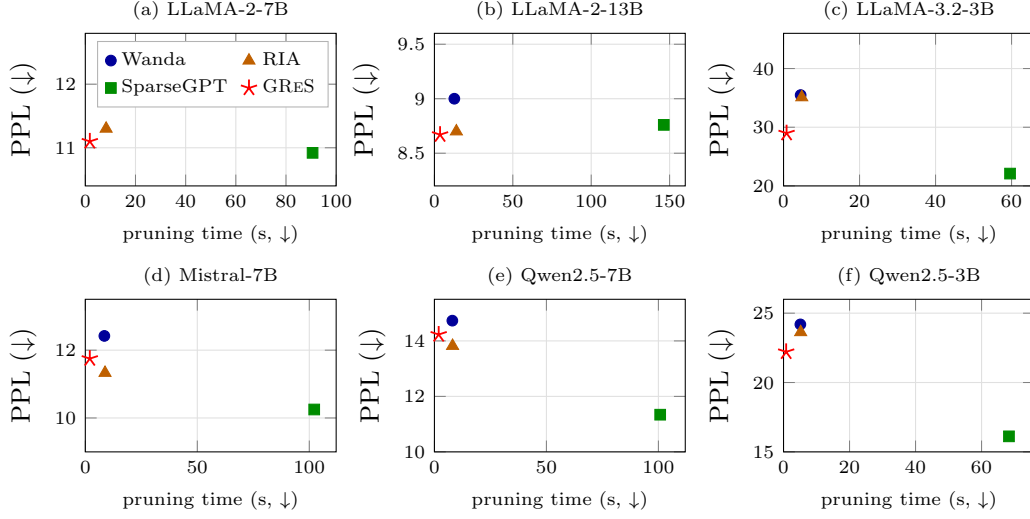


Figure S1: Quality–cost trade-off at 2:4 (PPL vs. pruning time, both ↓; lower-left is better) across all six models (full version of main-text Fig. 4). GRES has the lowest pruning-stage cost on every model while reaching lower perplexity than Wanda on all six and than RIA on four, approaching SparseGPT and matching it on LLaMA-2-13B.

Table S16: Component ablation at 4:8 (WikiText-2 PPL ↓), companion to main-text Table III. Components add cumulatively to Wanda, except that the two permutations are *alternatives* (GRES uses cheap v2); “+comp.” adds the block-OBS compensation.

Variant	LLaMA-2-7B	LLaMA-2-13B	LLaMA-3.2-3B	Mistral-7B	Qwen2.5-7B	Qwen2.5-3B
Wanda (diagonal)	8.61	7.04	21.27	8.41	10.39	15.38
+ off-diagonal (block Gram)	8.61	7.02	21.04	8.44	10.43	15.26
+ Gram shrinkage	8.61	7.02	21.04	8.44	10.43	15.26
+ spectral perm. (alt.)	8.43	6.95	20.44	8.40	10.48	15.10
+ cheap perm. (v2) = GRES	8.23	6.92	19.41	8.35	10.34	14.81
+ block-OBS comp. (= GRES+C)	8.32	6.91	19.06	8.34	10.37	14.76

Table S17: Composability of the off-diagonal Gram lift at 2:4 (WikiText-2 PPL ↓): a diagonal base (“w/o”: Wanda or RIA) vs. the same base with GRES’s block-Gram selection (“w/”); Δ is the change. On the Wanda base the lift helps on all six models; on the RIA base it is large where within-block correlations are strong (LLaMA-3.2-3B, Qwen2.5-3B) and roughly neutral where the RIA base is already strong.

Base		LLaMA-2-7B	LLaMA-2-13B	LLaMA-3.2-3B	Mistral-7B	Qwen2.5-7B	Qwen2.5-3B
Wanda	w/o	12.06	9.00	35.48	12.42	14.73	24.19
	w/	11.10	8.67	28.98	11.75	14.23	22.21
	Δ	−0.96	−0.33	−6.50	−0.67	−0.50	−1.98
RIA	w/o	11.30	8.70	35.10	11.33	13.82	23.63
	w/	11.10	8.67	28.98	11.75	14.23	22.21
	Δ	−0.20	−0.03	−6.12	+0.42	+0.41	−1.42

Supported models. LLaMA-2 (7B/13B/70B), LLaMA-3.2-3B, Mistral-7B, and Qwen2.5 (3B/7B)—any HuggingFace LLaMA-family model, with automatic grouped-query-attention detection.

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